

BIT 4107-Mobile Application Development

ENHANCED PROFESSIONAL STUDENT LOGBOOK

Student name: KELLY MAROA

Admission Number:BIT/2024/49929

School/Faculty: COMPUTING AND INFORMATICS

Programme:Blood_Donation_App

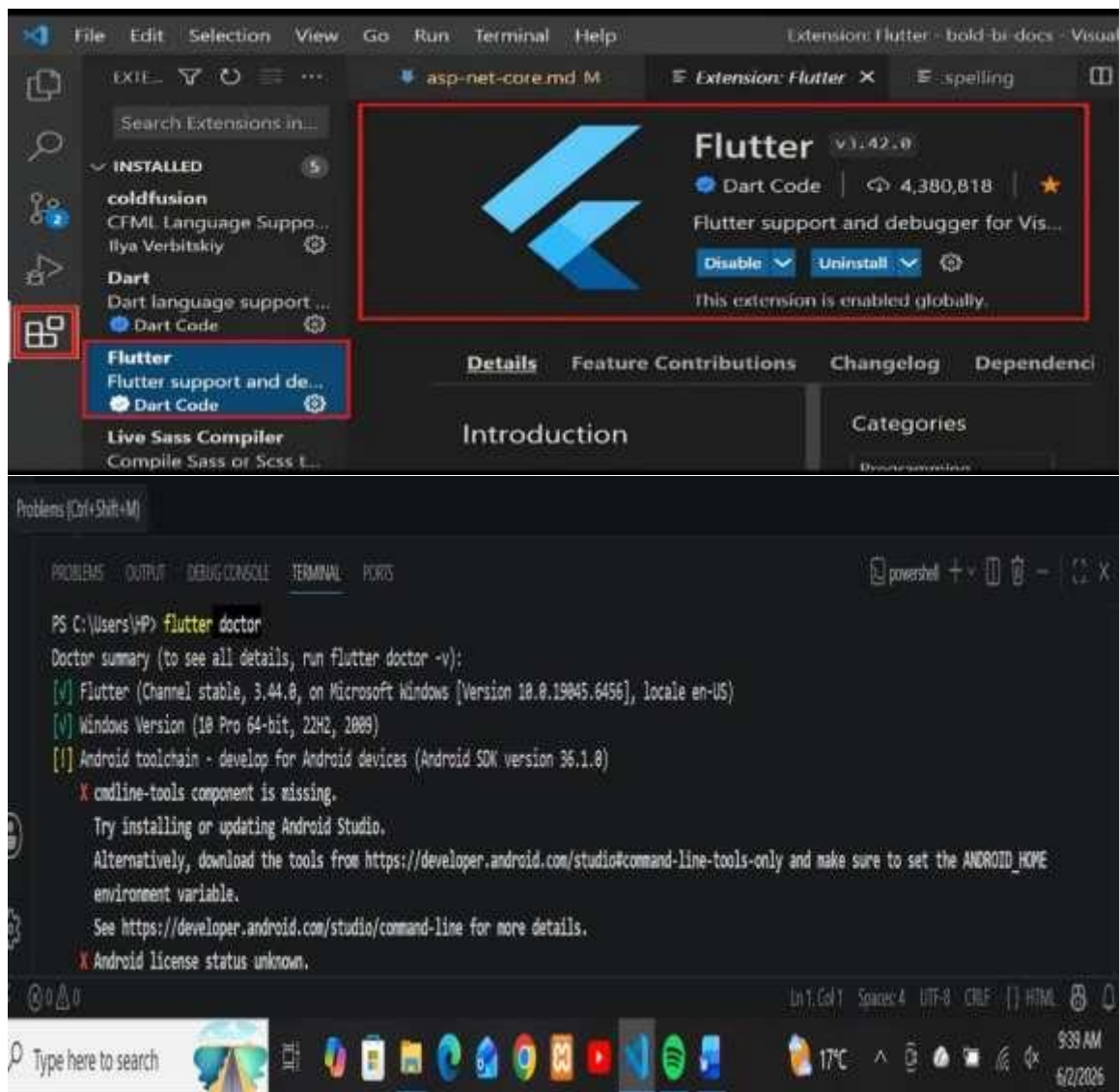
Academic Year: THREE

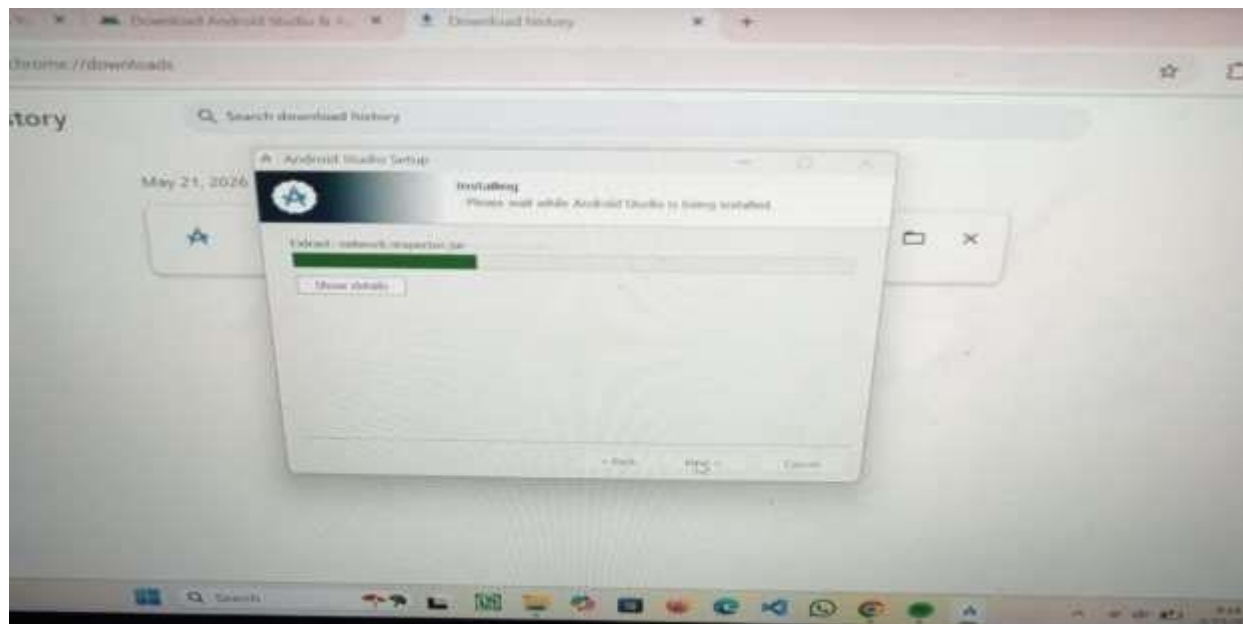
Semester:TWO

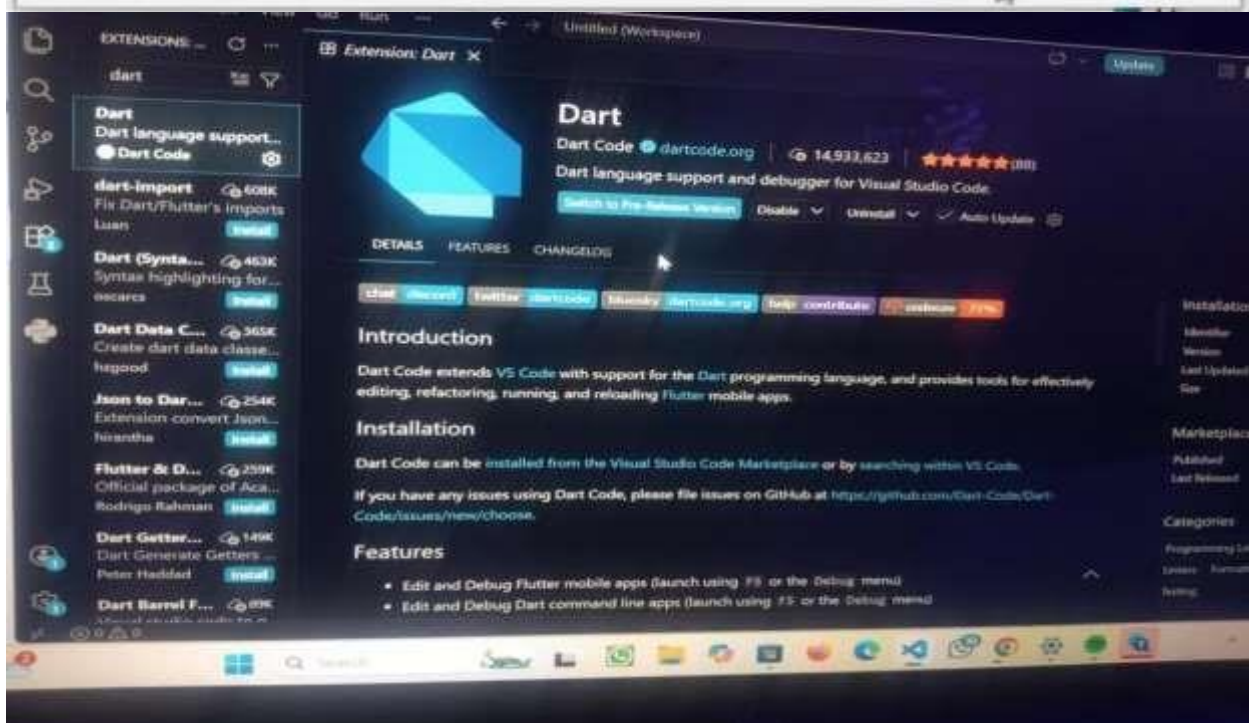
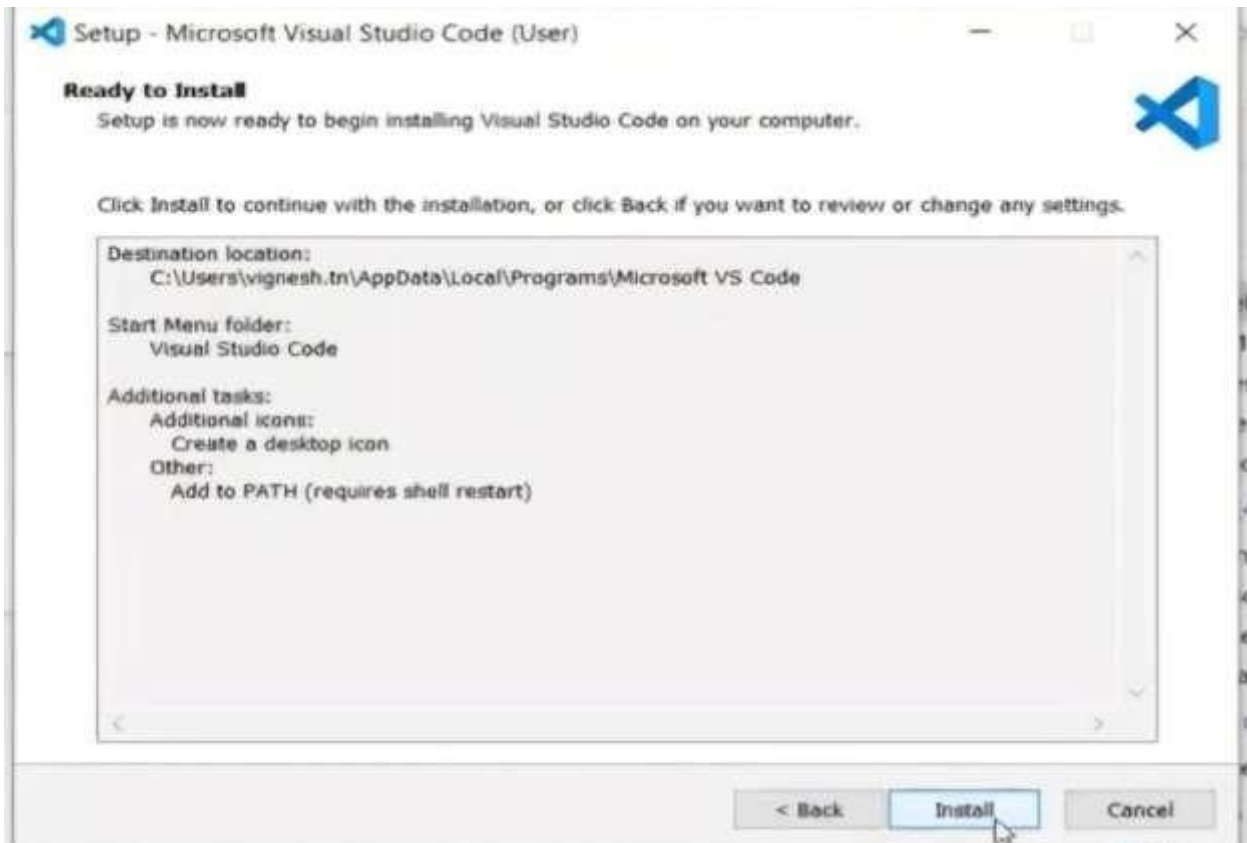
Lecturer: MR.MICHAEL NYORO

Phone Number:0725104634

WEEK 1: INTRODUCTION TO MOBILE APPLICATION DEVELOPMENT







TOOLSANDSOFTWAREUSED 1.Android
Studio

2.VS Code

3.Flutter SDK

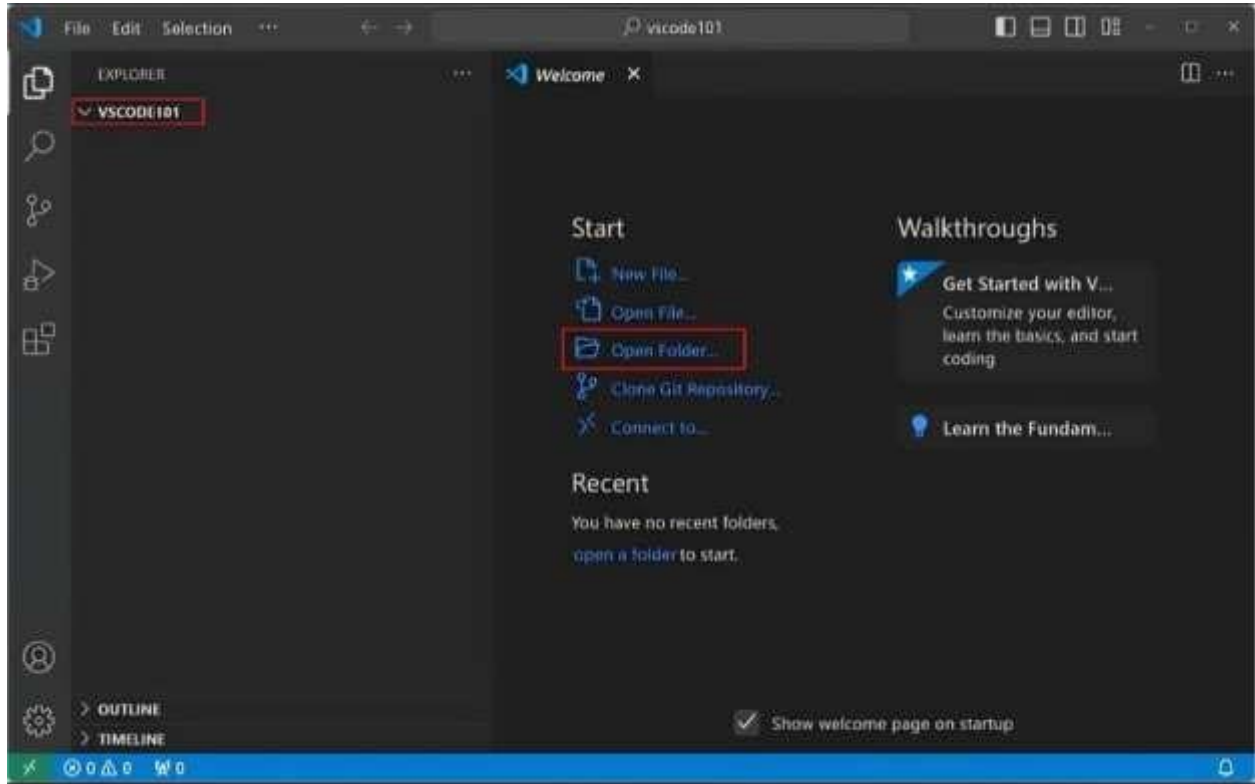
4.Dart

5.Android Phone

PERSONAL REFLECTION

Installing the tools was challenging at first due to errors and configuration issues, but it improved my problemsolving skills. I gained practical experience in setting up development environments and became more confident in using modern software tools for app development.

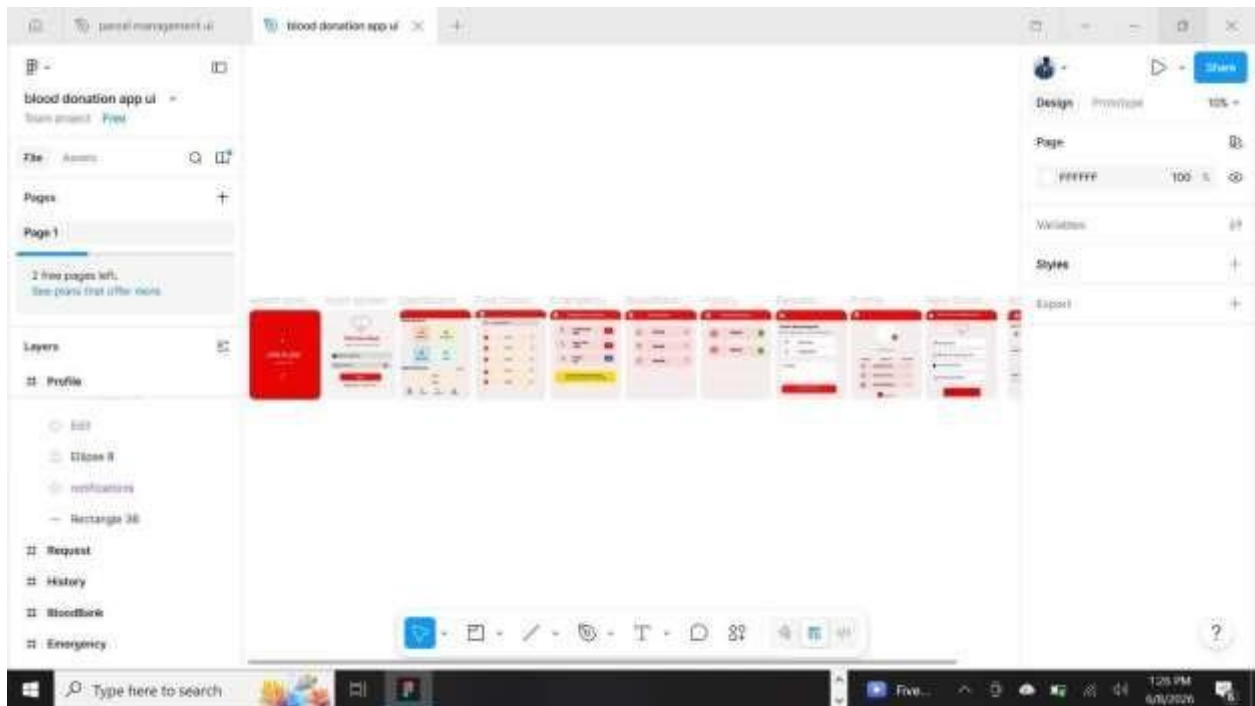
WEEK 2: MOBILE DEVELOPMENT ENVIRONMENT



PERSONAL REFLECTION

I set up the mobile development environment by installing and configuring tools like Android Studio, Flutter SDK, and device emulators. This process helped me understand how different components work together, though I faced minor setup challenges that enhanced my troubleshooting and technical skills.

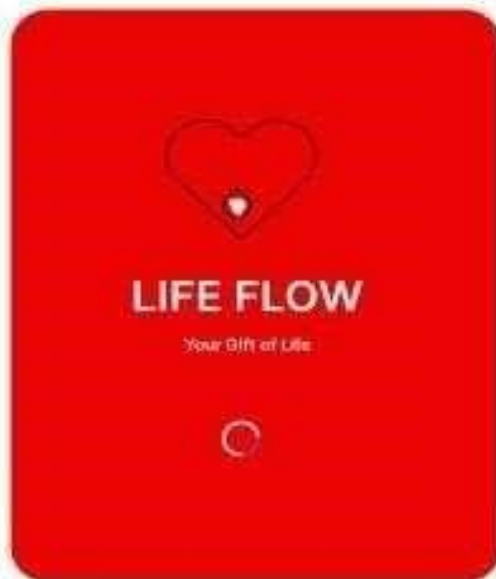
WEEK 3: USER INTERFACE DESIGN



login screen



plash screen

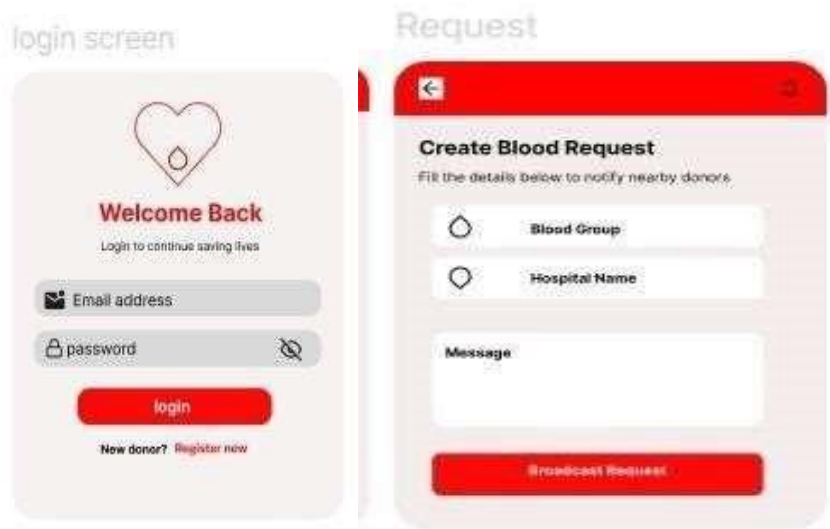


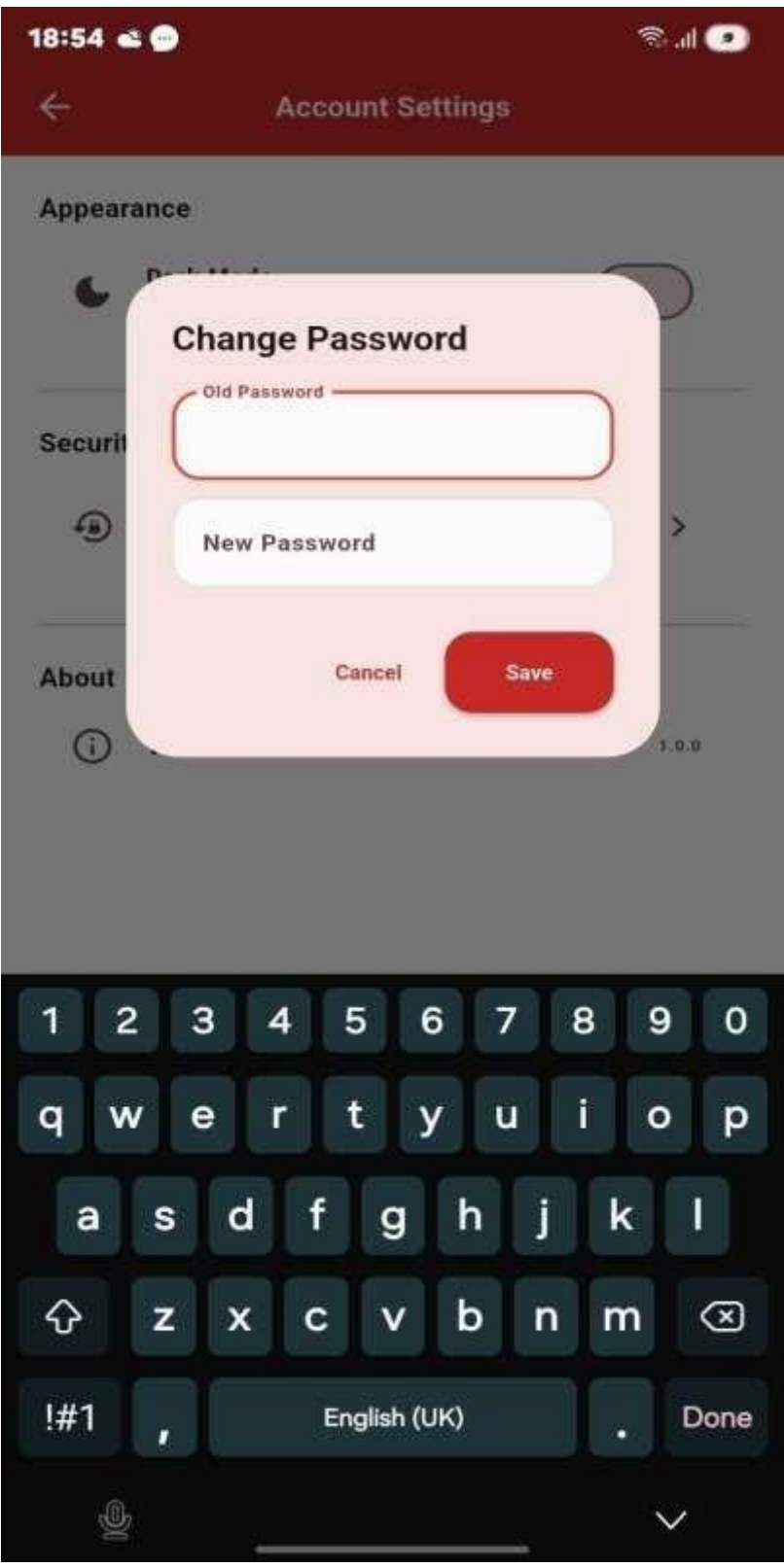


Personal reflection

I focused on user interface design for the blood donation app using Figma. I learned how to create and organize screens, design user-friendly layouts, and improve overall system usability. This improved my understanding of UI/UX principles and enhanced my design skills.

Week 4: Event handling and user interaction





Personal reflection

I learned about event handling and user interaction in the blood donation app. I implemented how users interact with the system through buttons, forms, and input fields. This improved my understanding of how applications respond to user actions and enhanced my programming skills.

Week 5: Data management

19:08



New Donor Registration



Full Name



Kelly maroa

Blood Group (e.g. O+)



O+

Location/City



Machakos

Phone Number



0785324646

Register as Donor



Search Blood Group



Anna Itotiah W.
Nairobi



Sharon Kendi.
Nakuru



Faith Mueni.
Mombasa



Roy Gichinga.
Kisumu



Michelle Njogu.
Eldoret



Kelley Maroa
Migori



Nicole Chepkemboi
Nandi



Home



Search



Request



Profile



Personal reflection

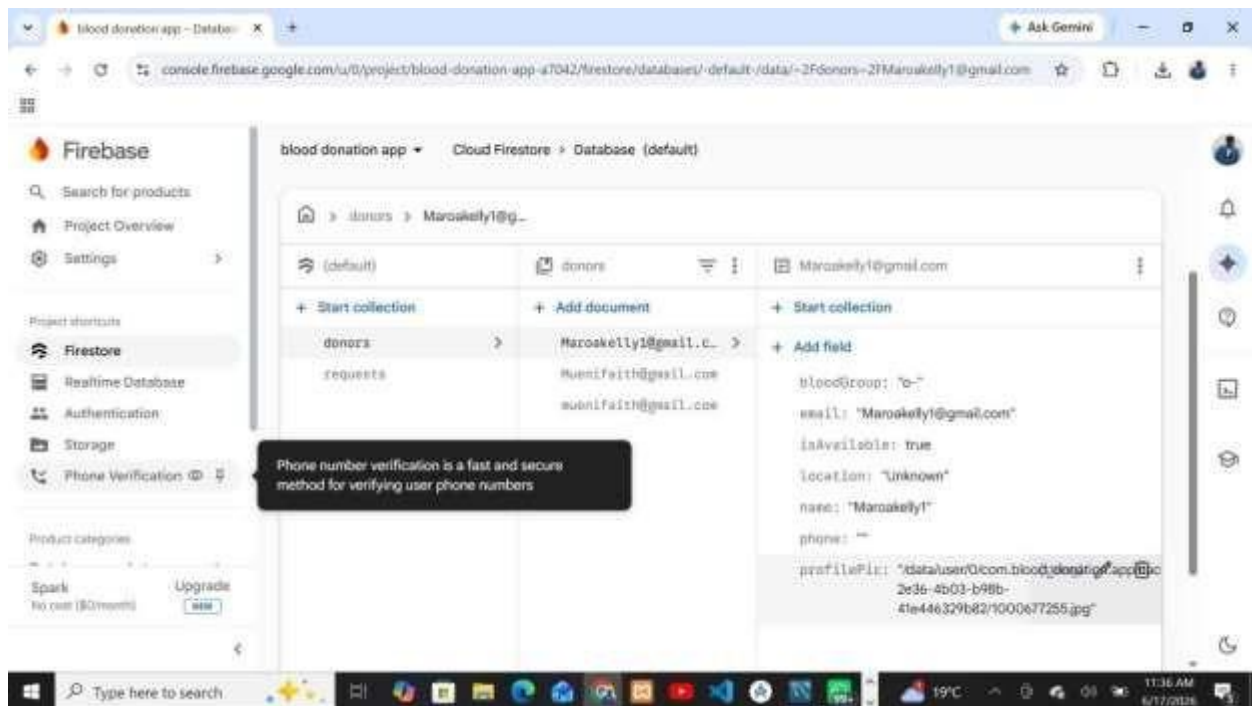
I focused on data management in the Blood Donation App. I learned how to store, retrieve, and update donor and request information using local storage and databases. This improved my understanding of how data is handled efficiently and enhanced my skills in managing application data.

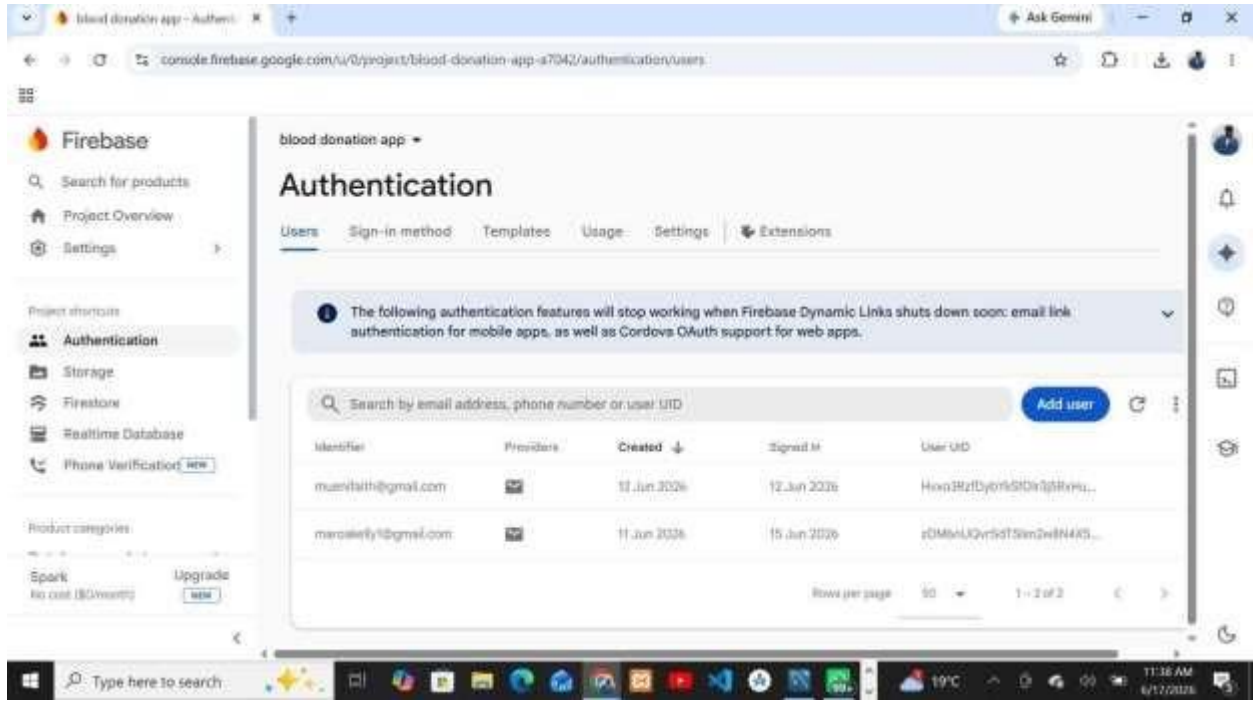
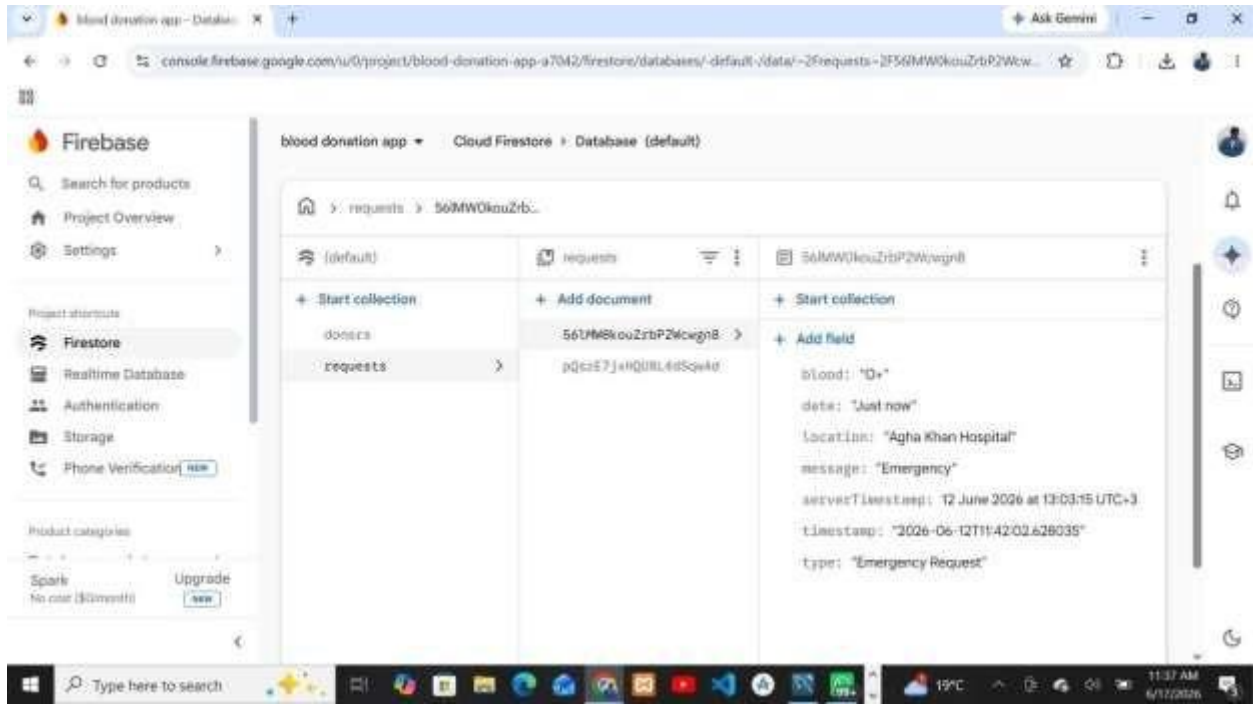
Week 6: Networking and APIs

```
1  {
2    "project_info": {
3      "project_number": "596793580149",
4      "project_id": "blood-donation-app-a7042",
5      "storage_bucket": "blood-donation-app-a7042.firebaseioapp"
6    },
7    "client": [
8      {
9        "client_info": {
10          "mobilesdk_app_id": "1:596793580149:android:d5f7eac4ef37ed8144363c",
11          "android_client_info": {
12            "package_name": "com.blood_donation.app"
13          }
14        },
15        "oauth_client": [],
16        "api_key": [
17          {
18            "current_key": "AIzaSyDVDy62fZd7rrAi_DwKwh7gyK9Nb91T9Wo"
19          }
20        ],
21        "services": {
22          "appinvite_service": {
23            "other_platform_oauth_client": []
24          }
25        }
26      }
27    ],
28    "configuration_version": "1"
29  }
```

A blood donation app requires networking to connect users with donors and hospitals, enable real-time updates, send requests, share location data, and store information on cloud servers for accessibility and communication.

Week 7: Data storage and Database integration in Mobile application





AB-	Agha Khan Request - 2026-06-08	
B+	Kenyatta hospital Request - 2026-06-05	
O+	Thika level 5 Request - 2026-06-05	



Search Blood Group



Anna Itotiah W.
Nairobi



Sharon Kendi.
Nakuru



Faith Mueni.
Mombasa



Roy Gichinga.
Kisumu



Michelle Njogu.
Eldoret



Kelley Maroa
Migori



Nicole Chepkemboi
Nandi



Home



Search



Request



Profile

Personal reflection

Learning about data storage and database integration helped me understand how mobile apps manage and persist user data efficiently. I now appreciate the importance of choosing the right database and ensuring secure, reliable data handling in application development.

WEEK 8: Handling user inputs and Events

20:39



Welcome Back

Login to continue saving lives



Email Address

example@mail.com



Password



Maroakelly2
@gmail.com

urgent

o



20:40



Account Settings

Appearance



Dark Mode



Change Password

Old Password

New Password

Cancel

Save

Security



About



Version

1.0.0



20:39



Welcome Back

Login to continue saving lives



Email Address

example@mail.com

Enter a valid email



Password



Password must be 6+ characters

Login

New donor? [Register Now](#)

20:40



Hello, Maroakelly1



 Nairobi, KE (Lat: -1.19)

Quick Services



Find Donor



Emergency



Blood Bank



History

Smart Match Suggestion



Best Match Found!

Maro Kelly (O+) is 500m away and available.



Urgent Requests

[View All](#)



Oasis
Just now
urgent!!



Home



Search



Request



Profile

20:40



Hello, Maroakelly1



Search Donor, Group or Location

mue



Faith Mueni

Machakos • AB-



muenifaith

Unknown • Unknown



mue

mueni

muen



20:45



← lib/main.dart



```
476         const SizedBox(height: 40),
477         Text(
478           "Welcome Back",
479           style: TextStyle(
480             fontSize: 32,
481             fontWeight: FontWeight.bold,
482             color: Colors.red.shade900,
483           ),
484         ),
485         const SizedBox(height: 10),
486         Text(
487           "Login to continue saving lives",
488           style: TextStyle(color: Colors
489             .grey.shade600, fontSize: 16),
490         ),
491         const SizedBox(height: 50),
492         TextFormField(
493           controller: emailController,
494           decoration: const InputDecoration(
495             labelText: "Email Address",
496             hintText: "example@mail.com",
497             prefixIcon: Icon(Icons
498               .email_outlined),
499             ),
500           keyboardType: TextInputType
501             .emailAddress,
502           validator: (v) => v!.contains('@') ?
503             null : "Enter a valid email",
504         ),
505         const SizedBox(height: 20),
506         TextFormField(
507           controller: passwordController,
508           obscureText: !isPasswordVisible,
509           decoration: InputDecoration(
510             labelText: "Password",
511             prefixIcon: const Icon(Icons
512               .lock_outline),
513             suffixIcon: IconButton(
514               icon: Icon(Icons
515                 .password_vizib1e),
```



Home



Inbox



Explore



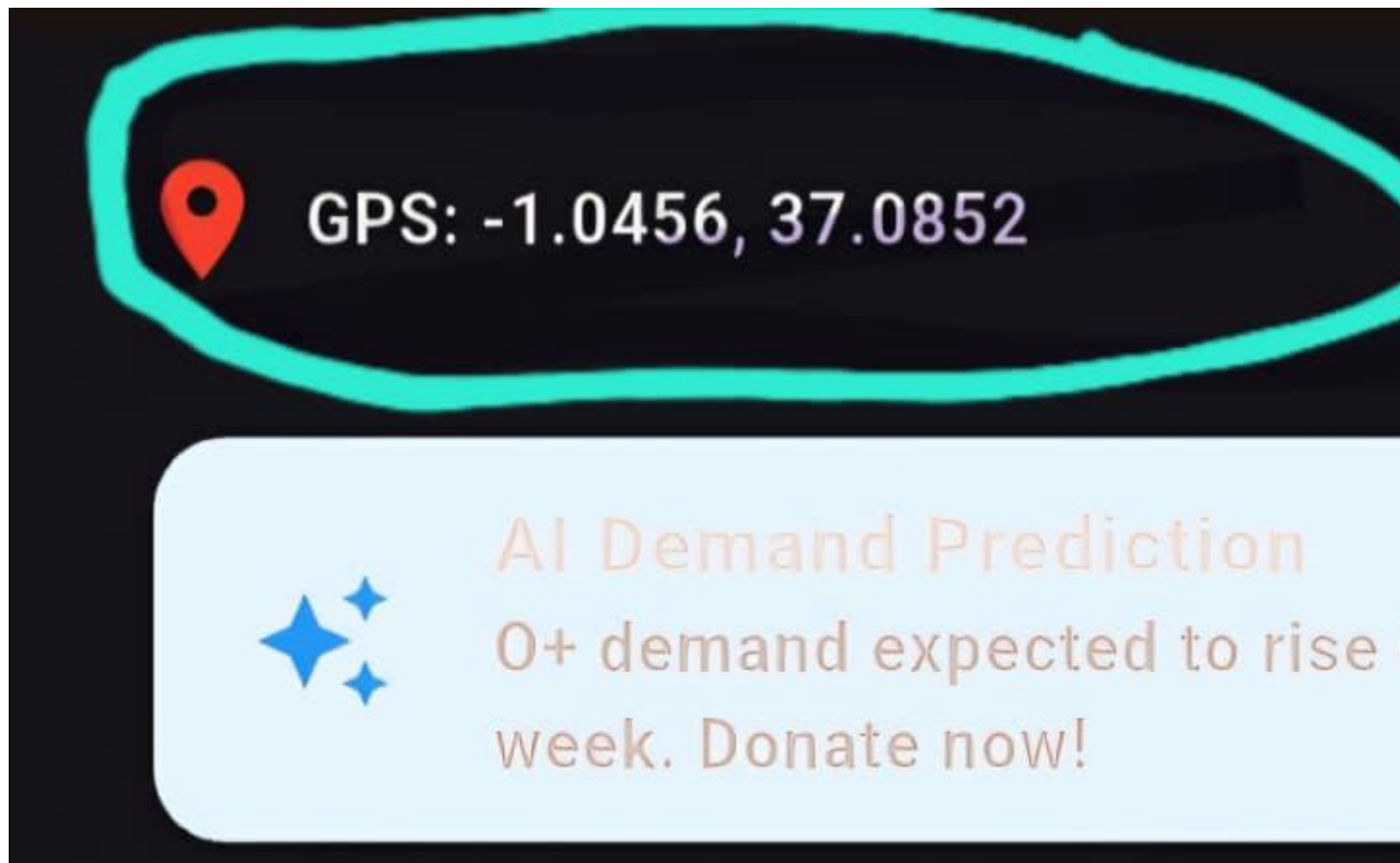
Copilot

Personal reflection

This topic helped me understand how mobile apps handle user input through events and gestures and how to do validation. I learned to structure functionality using classes, making applications more organized, interactive, and easier to maintain while improving user experience.

WEEK 9: DEVICE FEATURURES

INTEGRATION



Account Settings

Security



Change Password

Update your login security



Preferences

Dark Mode



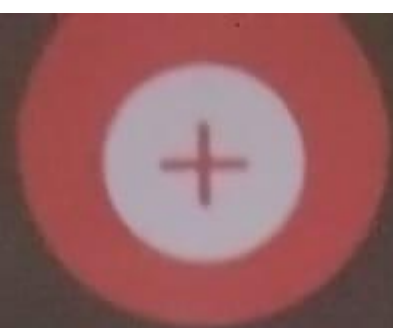
Biometric Security

Use Fingerprint or Face ID



App Version

2.1.0



Welcome B



Email Address

Fingerprint



blood_donation

Authentication re

Verify identity

PERSONAL REFLECTION

Through Week 9, I learned how to integrate device features like GPS and biometrics into my blood donation app. This improved my understanding of creating interactive, user-friendly solutions while considering permissions, privacy, and real-world application efficiency.

WEEK 10: COMPREHENSIVE APPLICATION
DEVELOPMENT INTEGRATION AND TESTING

20:21



LIFE FLOW

.

Account Settings

Security



Change Password

Update your login security



Preferences

Dark Mode



Biometric Security

Use Fingerprint or Face ID



App Version

2.1.0

20:03



Life Flow

Maroakelly1

Available to Donate



Accepting Requests

Toggle availability for blood requests



Donor ID



Badge



First Gift



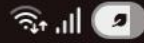
Donor ID Card



Settings



20:21



Life Flow



Maroakelly1

Available to Donate



Accepting Requests

Toggle availability for blood requests



Donation Eligibility

Eligible to Donate

3

Donated

0

Points

9

Lives Saved

Badges



First Gift



Saver



Verified



Donor ID Card



20:20 [icons]



Life Flow

Create Blood Request

Blood Group Needed



Hospital / Collection Point

Additional Message

Broadcast Request

Broadcast Successful!



PERSONAL REFLECTION

In Week 10, I learned how to integrate all the different components of a mobile application into one complete system. I improved my understanding of testing, debugging, and enhancing user experience. This helped

me appreciate the importance of building a functional, reliable, and user-friendly application.

WEEK 11:APPLICATION DEPLOYEMENT AND PUBLISHING

19:56



My projects 🤔



Search





LIFE FLOW

.

BANNER



BLOOD CONNECT

DONATE BLOOD. SAVE LIVES.

Blood Connect bridges the gap between donors and those in need. Together, we can build a healthier and stronger community.



Find Donors



Request Blood



Nearby Donors

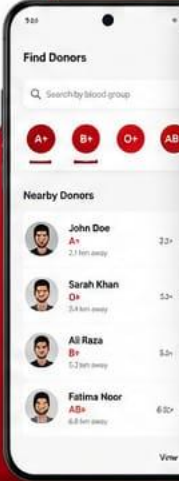


Instant Notifications



Secure & Reliable

YOUR DONATION CAN SAVE A LIFE





BE A HERO. **DONATE BLOOD.**

TOGETHER, WE SAVE MORE LIVES.

POWERED BY  **Firebase**

BLOOD CONNECT – USER GUIDE

A simple guide to help you navigate and make the most of the Blood Connect app.

1 REGISTER / LOGIN



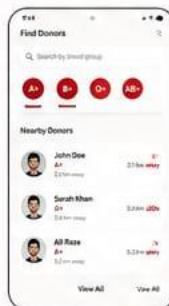
Create a new account or login using your credentials.

2 DASHBOARD



Access all features from the dashboard. Choose what you need.

3 FIND DONORS



Search donors by blood group and location. Contact them directly.

4 REQUEST BLOOD



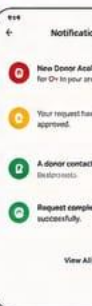
Fill in the details to request blood in emergencies.

5 MY REQUESTS



Track the status of your blood requests in real time.

6 NOTIFICATIONS



Stay updated with time alerts and notifications.



Your data is safe with us. Blood Connect uses Firebase for secure authentication and real-time data synchronization.



Firebase



Authentication

ONE PAGE USER GUIDE

BLOOD CONNECT APP – USER GUIDE

1. Introduction

The Blood Connect App is designed to help users find blood donors quickly and request blood during emergencies. The application provides a simple and user-friendly interface for both donors and recipients.

2. Getting Started

- Download and install the app.
- Open the app and register using your email and password.
- Log in to access the dashboard.

3. Key Features

- Search Donors: Users can search for blood donors based on blood group and location.
- Request Blood: Users can submit emergency blood requests.
- Donor Registration: Users can register as blood donors.

- Profile Management: Users can update their personal details.

4. How to Use the App

- After logging in, navigate to the dashboard.
- Select “Search Donors” to find available donors.
- Use filters such as blood group to refine results.
- Click “Request Blood” to send an emergency request.
- Fill in the required details and submit.

5. Notifications

Users receive alerts when:

- A blood request is successfully submitted
- A donor responds to a request

6. Safety and Privacy

- User data is securely stored.
- Personal information is only used for donation purposes.

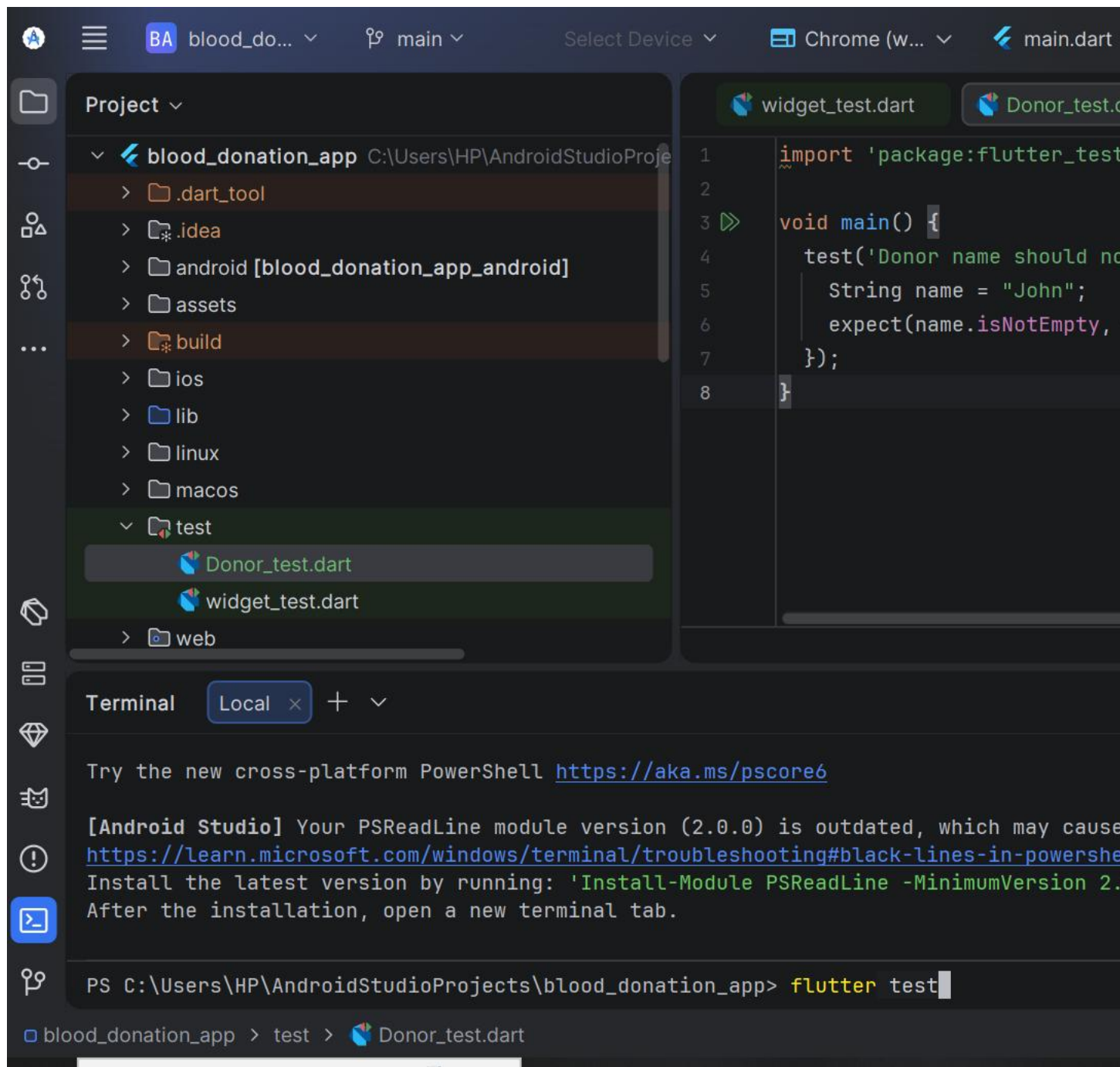
7. Conclusion

The Blood Connect App provides a fast, reliable, and efficient way to connect blood donors with recipients, helping save lives through technology.

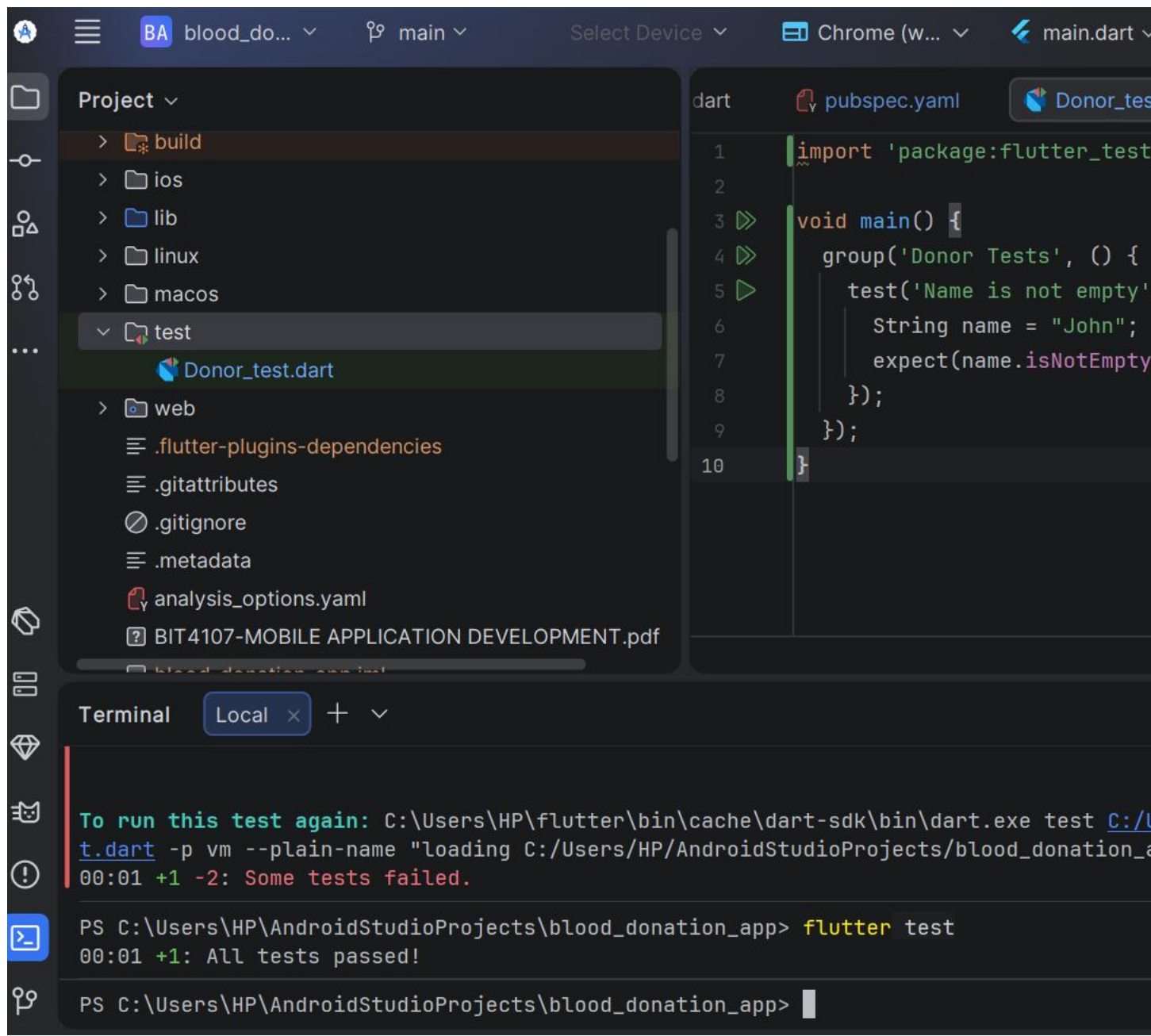
PERSONAL REFLECTIONS

Week 11 enhanced my understanding of application deployment. I learned to generate signed builds, prepare store assets, and perform final testing. It showed me that presentation, documentation, and quality assurance are essential before releasing an app to real users.

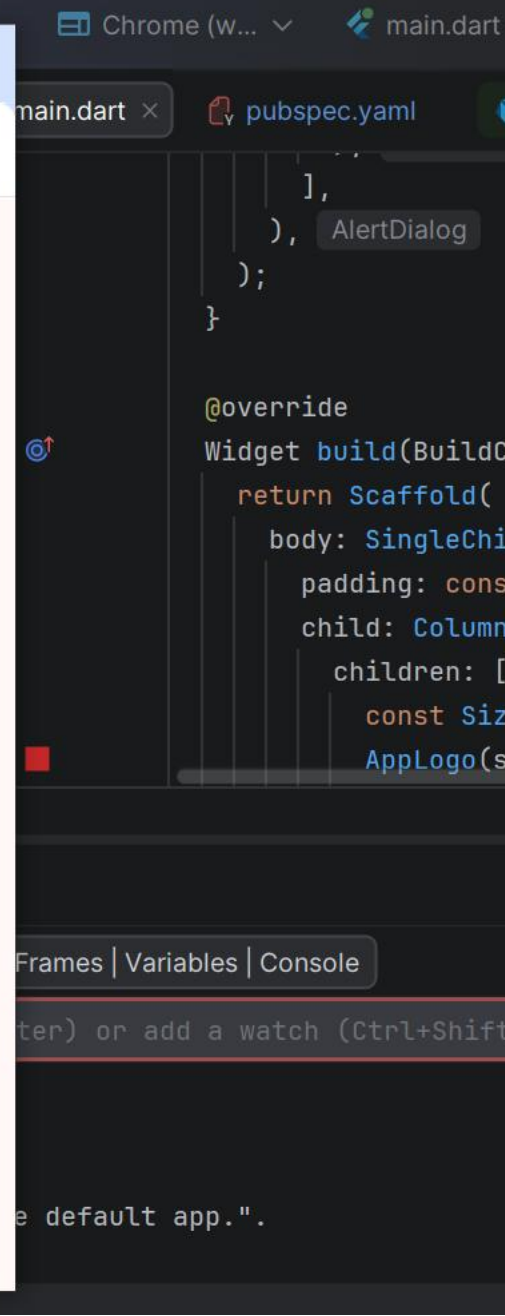
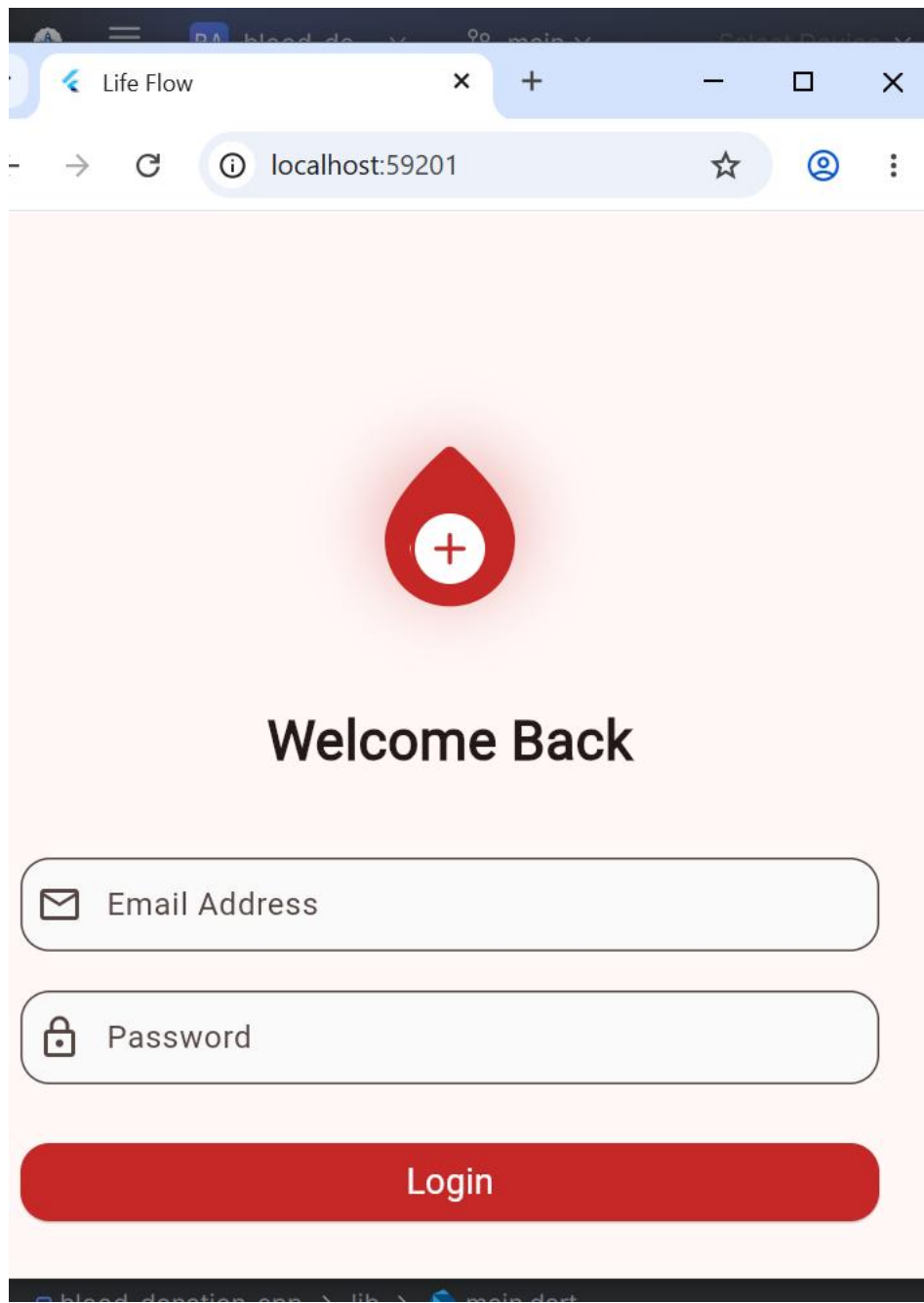
WEEK 12: TESTING AND DEBUGGING MOBILE APPLICATIONS



Unit testing passed



Debug on chrome android studio



Validation



Welcome Back



Email Address

yyggh



Password

..|

Login



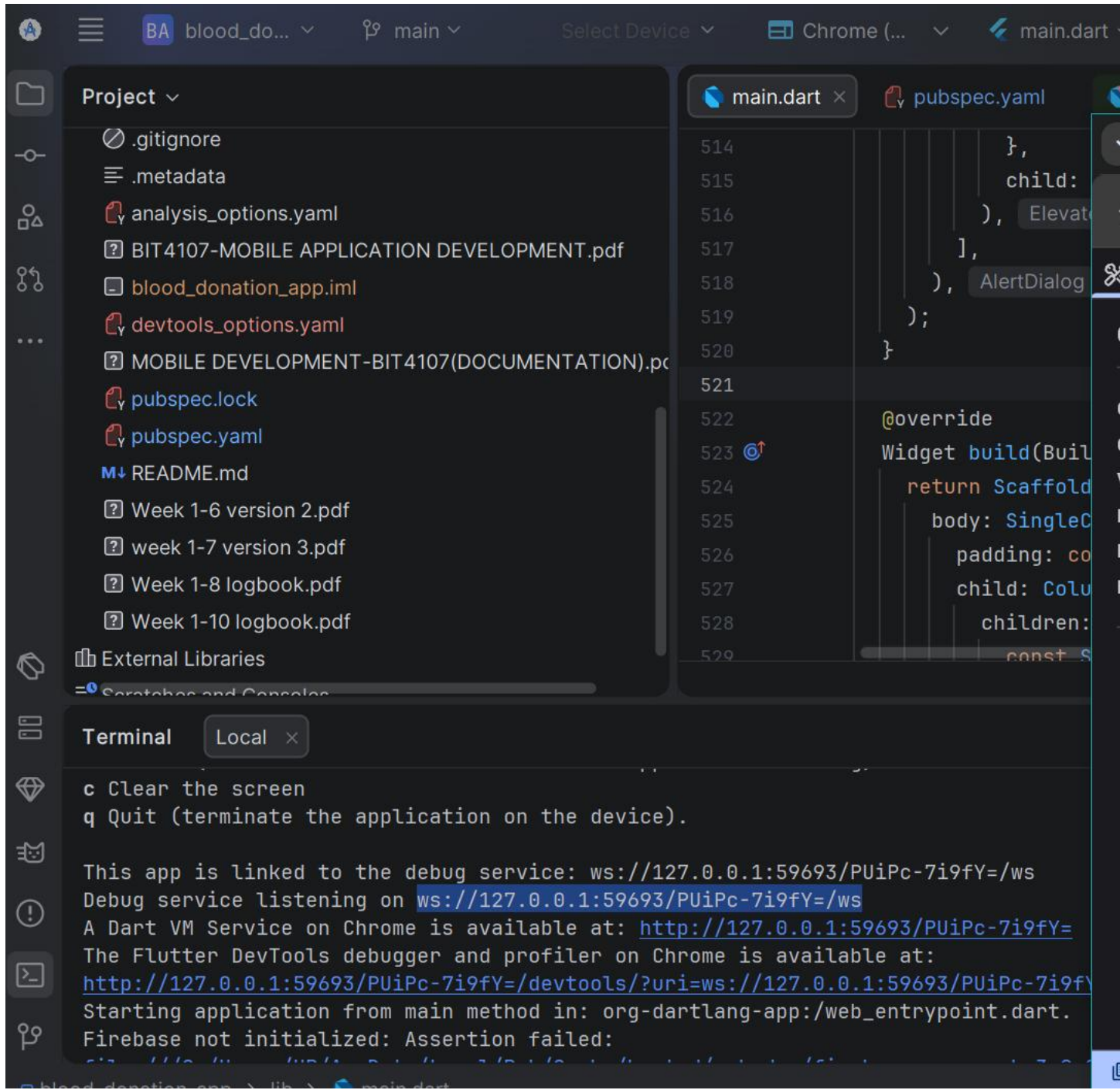
The email address is badly formatted.

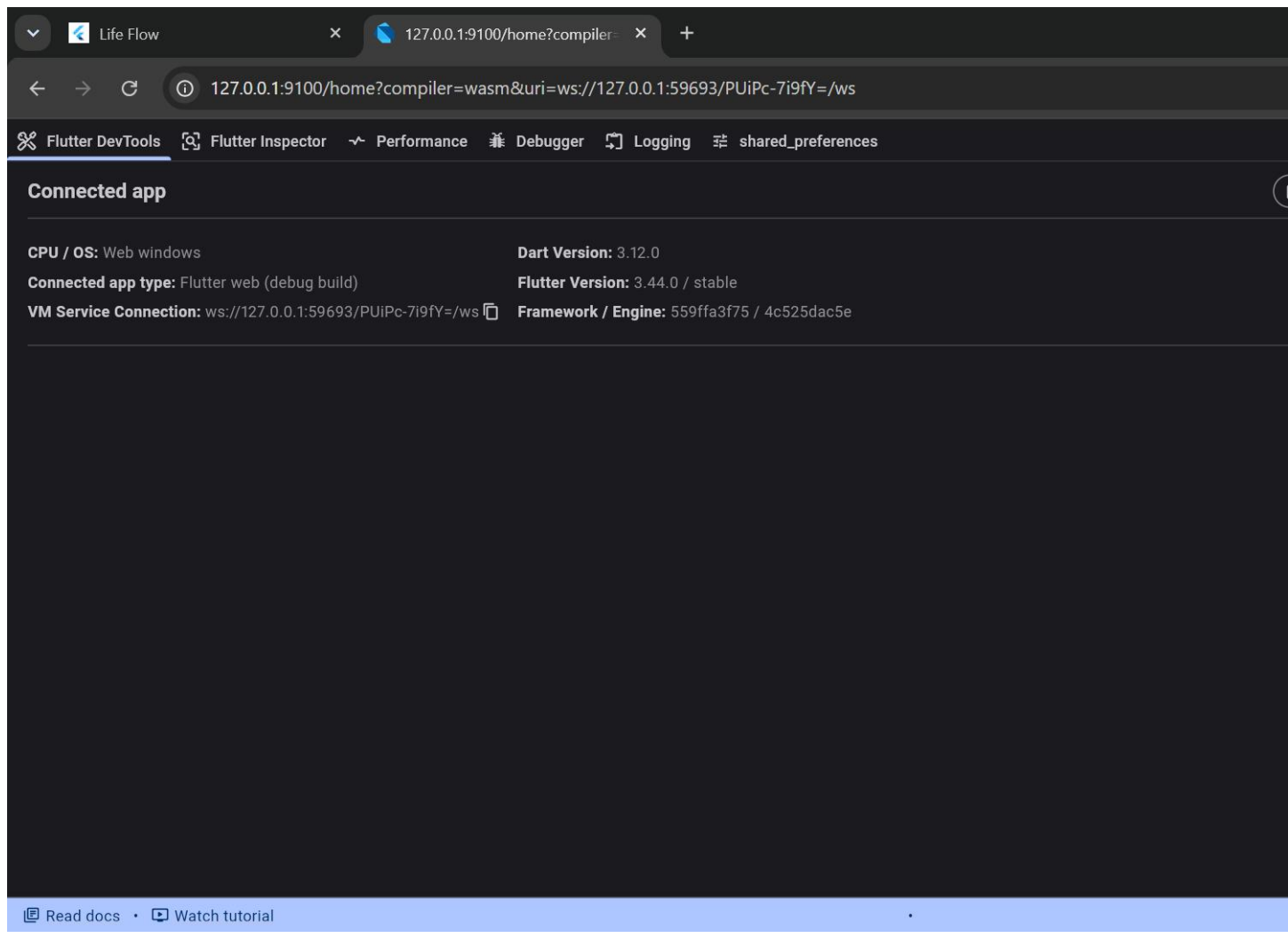


Personal reflection

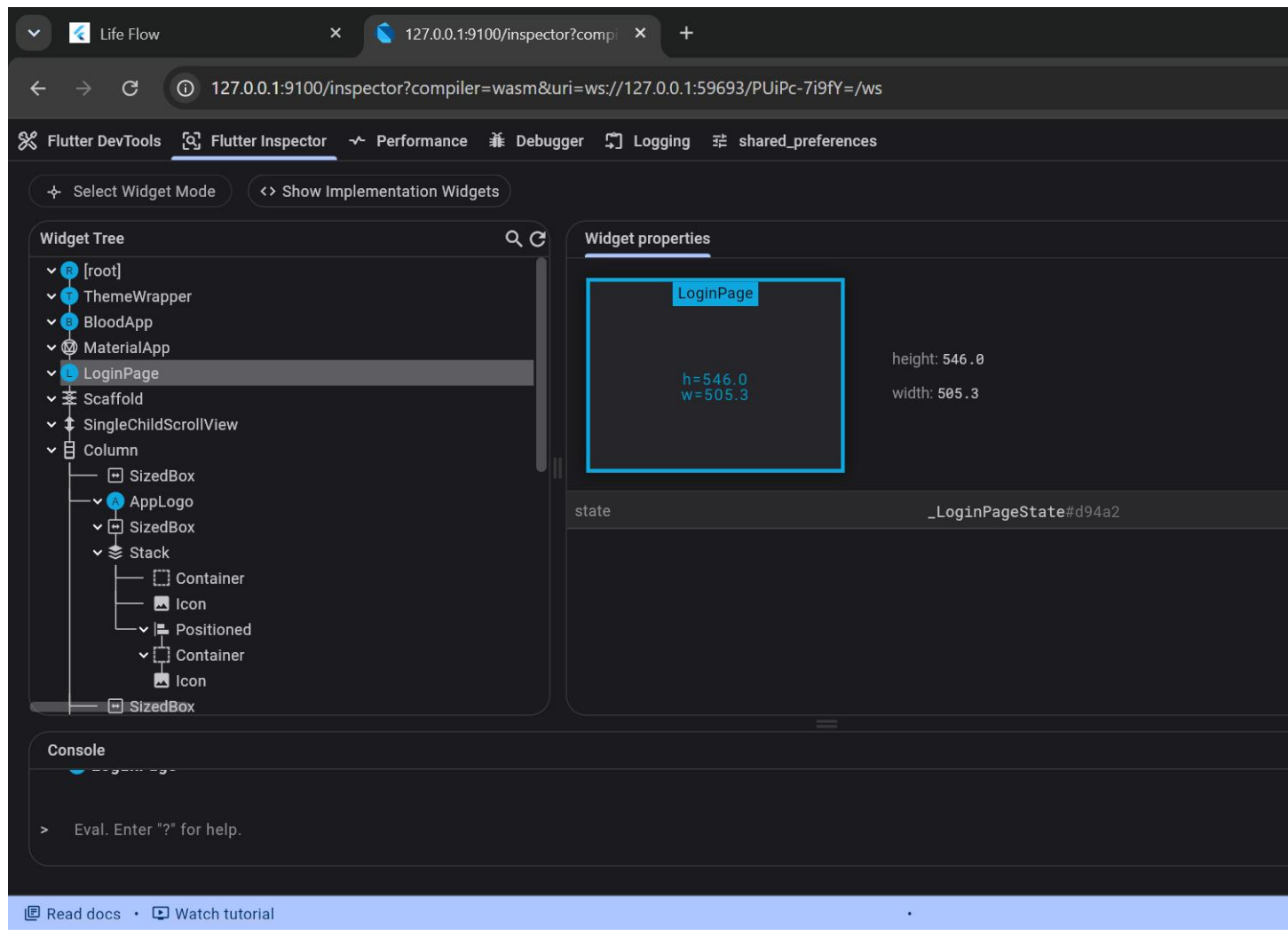
I improved my understanding of Flutter DevTools and debugging. Faced issues connecting tools but learned how to navigate Inspector, Debugger, and Chrome DevTools. Gained confidence troubleshooting errors independently.

WEEK 13:PERFOMANCE OPTIMIZATION

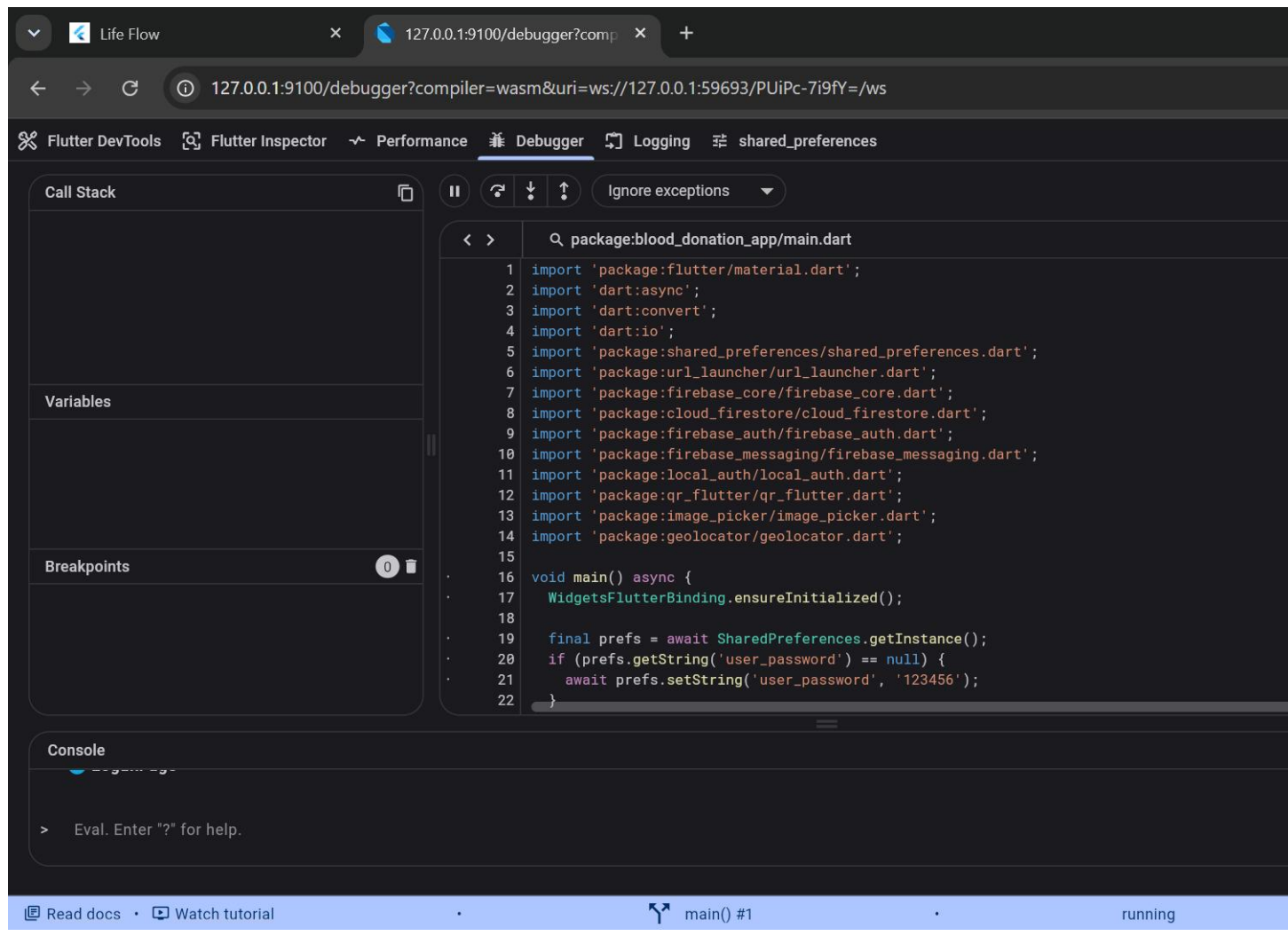




Flutter inspector



debugger



perfomance

Life Flow

127.0.0.1:9100/performance?co

127.0.0.1:9100/performance?compiler=wasm&uri=ws://127.0.0.1:59693/PUiPc-7i9fY=/ws

Flutter DevToolsFlutter InspectorPerformance

How to use Chrome DevTools for performance profiling

The Flutter framework emits timeline events as it works to build frames, draw scenes, and track other activity such as garbage collections. These events are exposed in the Chrome DevTools performance panel for debugging.

You can also emit your own timeline events using the dart:developer [Timeline](#) and [TimelineTask](#) APIs for further performance analysis.

Optional flags to enhance tracing

- `debugProfileBuildsEnabled`: Adds Timeline events for every Widget built.
- `debugProfileBuildsEnabledUserWidgets`: Adds Timeline events for every user-created Widget built.
- `debugProfileLayoutsEnabled`: Adds Timeline events for every RenderObject layout.
- `debugProfilePaintsEnabled`: Adds Timeline events for every RenderObject painted.

Instructions

- [Optional]* Set any desired tracing flags to true from your app's main method.
- Run your Flutter web app in [profile mode](#).
- Open up the [Chrome DevTools' Performance panel](#) for your application, and start recording to capture timeline events.

Read docs · Watch tutorial

web app

ElementsConsoleSources

MemoryDim 3rd parties

Interaction to Next Paint (INP)

1,192 ms

Your local INP value of 1,192 ms is poor.

[INP interaction pointer](#)

[Learn more about local and field metrics](#)

InteractionsLayout shifts

keyboardinput.flt-text-editi

pointerflutter-view

pointerflutter-view

Next steps

Field metrics

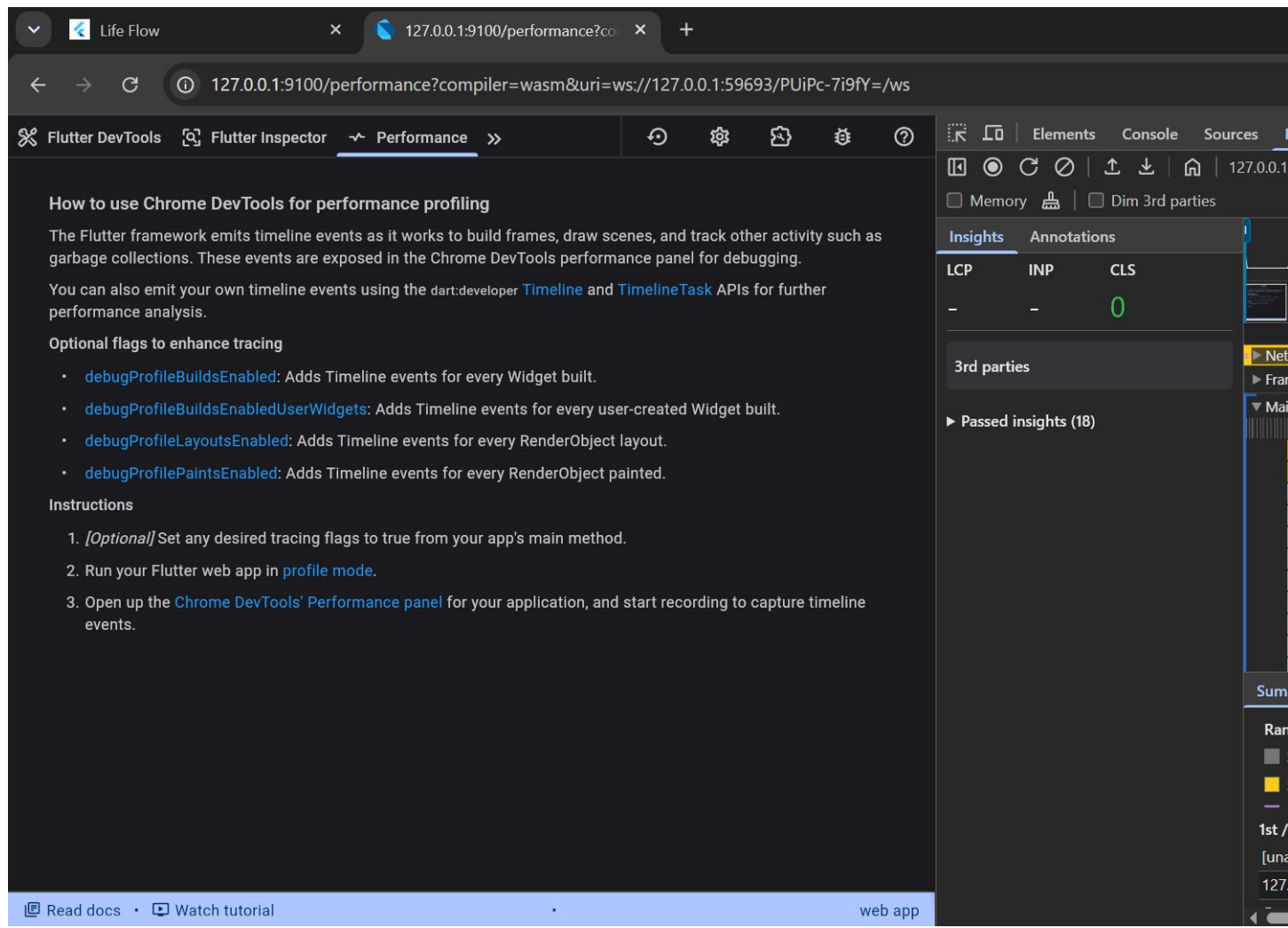
See how your local metrics compare to real us

Set up

Environment settings

Use the devtools to help and improve the app

Perfomance graph



Personal reflection

Focused on performance profiling and optimizing my Flutter web app. Learned to use Chrome DevTools effectively. Began identifying slow UI components and improved overall app responsiveness.

WEEK 14: FINAL CAPSTONE ASSIGNMENT

Enhanced problem-solving skills and workflow efficiency. Became more comfortable with debugging, profiling, and refining features. Reflected on progress and prepared for final improvements and project completion.